

MELBOURNE VIEWBANK, AustraliaWMO: **958740**Lat: **37.7408S**Lon: **145.0972E**Elev: **218**StdP: **14.58**Time Zone: **10.00 (AUS)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.5	37.9	32.3	27.0	52.8	34.1	29.0	50.9	21.4	55.9	19.6	54.7	3.5	40	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	95.7	66.8	91.2	66.0	87.0	65.1	70.8	83.6	69.0	82.7	67.4	80.6	10.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.5	101.8	72.4	65.5	94.9	71.4	63.6	88.8	70.8	34.8	83.8	33.2	83.3	32.0	80.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.0	15.8	31.2	105.3	1.4	4.3	30.2	108.4	29.4	111.0	28.5	113.4	27.5	116.6		
(5)			31.1	73.9	1.4	2.1	30.1	75.4	29.2	76.6	28.4	77.7	27.4	79.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	69.4	68.7	66.0	60.2	54.6	50.4	50.0	50.9	54.2	58.0	62.6	65.7	(6)	(7)
	DBStd	8.90	7.28	6.36	6.35	5.36	4.52	3.78	3.64	4.07	4.89	5.85	6.89	6.88	(7)	(8)
	HDD50	143	0	0	0	1	9	40	46	36	9	2	0	0	(8)	(9)
	HDD65	2711	29	29	65	163	322	437	466	436	328	233	130	73	(9)	(10)
	CDD50	3489	601	524	496	306	152	52	45	65	135	249	377	487	(10)	(11)
	CDD65	584	165	133	96	19	1	0	0	0	4	15	56	95	(11)	(12)
	CDH74	6767	1926	1359	973	202	3	0	0	0	23	277	765	1239	(12)	(13)
	CDH80	2978	990	603	390	35	0	0	0	0	2	75	301	582	(13)	(14)
(14) Wind	WSAvg	7.1	7.9	7.3	6.8	6.0	6.5	6.1	7.3	7.3	7.4	7.7	7.7	7.7	(14)	(15)
(15) Precipitation	PrecAvg	30.2	1.8	2.3	1.8	2.6	2.5	2.6	2.7	2.6	3.1	2.7	2.9	2.6	(15)	(16)
	PrecMax	42.7	4.3	7.2	3.6	6.3	5.6	5.4	4.3	4.4	5.9	6.3	6.6	6.7	(16)	(17)
	PrecMin	18.9	0.0	0.2	0.6	0.8	0.6	0.7	1.0	1.0	0.9	0.5	1.1	0.3	(17)	(18)
	PrecStd	6.1	1.0	1.7	0.9	1.4	1.1	1.0	0.9	1.0	1.3	1.5	1.4	1.6	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.7	97.8	95.8	84.2	73.2	64.6	63.3	68.4	77.0	87.3	93.3	100.5	(19)	(20)
		MCWB	68.7	67.3	66.7	62.3	57.7	54.9	52.9	54.9	58.7	62.0	65.3	65.7	(20)	(21)
	2%	DB	96.4	92.2	89.2	79.5	68.8	61.2	60.3	64.7	71.5	81.2	88.2	92.9	(21)	(22)
		MCWB	67.0	66.9	65.2	61.0	57.1	53.5	52.1	53.1	57.1	60.4	63.8	65.7	(22)	(23)
	5%	DB	90.8	87.6	84.0	75.6	66.0	59.1	58.3	61.5	67.9	76.3	82.9	86.4	(23)	(24)
		MCWB	66.7	66.3	64.2	60.1	56.1	52.7	50.9	51.7	55.0	59.1	63.3	64.4	(24)	(25)
	10%	DB	84.8	82.7	79.1	71.7	63.5	57.2	56.4	58.8	64.5	71.6	77.5	80.7	(25)	(26)
		MCWB	65.7	65.6	63.1	58.9	55.0	51.7	50.0	50.6	53.1	57.2	61.6	63.4	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.7	72.6	70.1	64.9	61.6	57.0	55.8	56.3	60.5	65.0	69.6	71.1	(27)	(28)
		MCDB	86.6	85.8	83.2	76.5	66.9	60.5	59.8	64.7	71.3	76.7	82.6	83.3	(28)	(29)
	2%	WB	71.2	70.2	68.0	63.1	58.7	55.0	53.7	54.3	58.1	62.4	66.9	68.2	(29)	(30)
		MCDB	83.9	83.4	78.2	72.8	65.2	58.6	57.7	61.9	68.6	75.8	79.5	82.8	(30)	(31)
	5%	WB	69.2	68.3	66.6	61.7	57.1	53.8	52.1	52.8	56.3	60.3	65.1	66.4	(31)	(32)
		MCDB	84.4	81.1	77.6	71.0	64.1	57.7	56.1	59.4	65.6	72.5	77.6	80.3	(32)	(33)
	10%	WB	67.2	66.8	64.9	60.2	55.7	52.5	50.8	51.4	54.4	58.3	62.9	64.7	(33)	(34)
		MCDB	82.1	79.0	76.6	69.3	61.9	56.4	55.1	57.3	62.4	69.5	75.0	78.0	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	20.7	19.8	18.5	14.5	12.6	11.8	13.7	16.4	19.7	20.5	21.6	(35)	(36)
		MCDBR	34.3	31.3	29.9	26.3	18.9	16.0	13.8	17.7	22.7	30.3	31.9	33.9	(36)	(37)
	5% WB	MCWBR	10.0	9.9	10.0	10.7	9.8	9.9	8.1	9.2	11.0	12.8	11.3	10.9	(37)	(38)
		MCWBR	26.3	24.7	22.8	21.4	15.6	13.6	12.0	15.4	19.6	25.2	25.3	26.1	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.350	0.356	0.332	0.330	0.310	0.302	0.302	0.313	0.336	0.337	0.347	0.340	(39)	(40)
		taud	2.509	2.499	2.564	2.540	2.582	2.591	2.582	2.542	2.469	2.491	2.489	2.528	(40)	(41)
	Ebn at Noon		312	302	298	278	265	258	265	280	291	305	311	316	(41)	(42)
		Edh at Noon	36	34	30	27	22	20	22	26	32	34	36	35	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		2271	1981	1584	1098	721	559	623	882	1259	1690	1997	2234	(43)	(44)
	RadStd		142	106	90	66	45	47	36	62	94	140	119	126	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.94	N/A	N/A	N/A	+1.53	+2.66	N/A	-185	+378	+203
(47) Regional (40 neighbors)		+0.92	N/A	N/A	+1.53	N/A	N/A	-32	-229	+311	+106

Nomenclature: See separate page